

CASE RECORDS of the MASSACHUSETTS GENERAL HOSPITAL

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Case 7-2026: A 91-Year-Old Man with Left Knee Pain

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PRESENTATION OF CASE

Dr. Erin H. Yang (Medicine): A 91-year-old man was evaluated in the orthopedic surgery clinic of this hospital because of left knee pain.

The patient had been in his usual state of health, residing in a senior living community and independent in activities of daily living, until 6 months before the current presentation, when left knee pain developed. He had undergone left total knee arthroplasty 20 years earlier, after which knee pain had not recurred until this presentation. The pain occurred at rest and was exacerbated by weight-bearing activity. During the next 4 months, the left knee pain worsened in severity, and the patient felt unstable when walking; he began using a rolling walker in his apartment and a wheelchair for longer distances.

Three weeks before the current presentation, the patient was evaluated at an orthopedic surgery clinic affiliated with another hospital. He reported that pain occurred throughout the left knee but was worst over the medial aspect. The patient had noticed swelling around the knee, with no redness or warmth. On examination, a moderate effusion was present in the knee; range of motion was normal, and there was tenderness on palpation of the medial joint line. Overall strength and sensation were normal, aside from a chronic right foot drop. He had pitting edema in the legs and diffusely dry skin, with no skin breakdown or ulcers.

Dr. Tina Shiang: Radiography of the left knee (Fig. 1A) showed total knee arthroplasty with no hardware complications. Cortical erosions, osteolysis, and osseous fragmentation were present at the lateral aspect of the distal femoral metadiaphysis, and there was a joint effusion. Additional imaging was recommended.

Dr. Yang: The erythrocyte sedimentation rate was 13 mm per hour (reference range, 0 to 20), and the blood level of C-reactive protein was 15.9 mg per liter (reference range, 0.0 to 7.9). Synovial fluid was aspirated from the left knee. The fluid was bloody, with a total nucleated-cell count of 6190 per microliter, of which 68% were neutrophils, 4% lymphocytes, and 28% macrophages. An immunoassay of synovial fluid for α -defensin was positive. Bacterial cultures of synovial fluid showed no growth after 2 days of incubation.

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CME



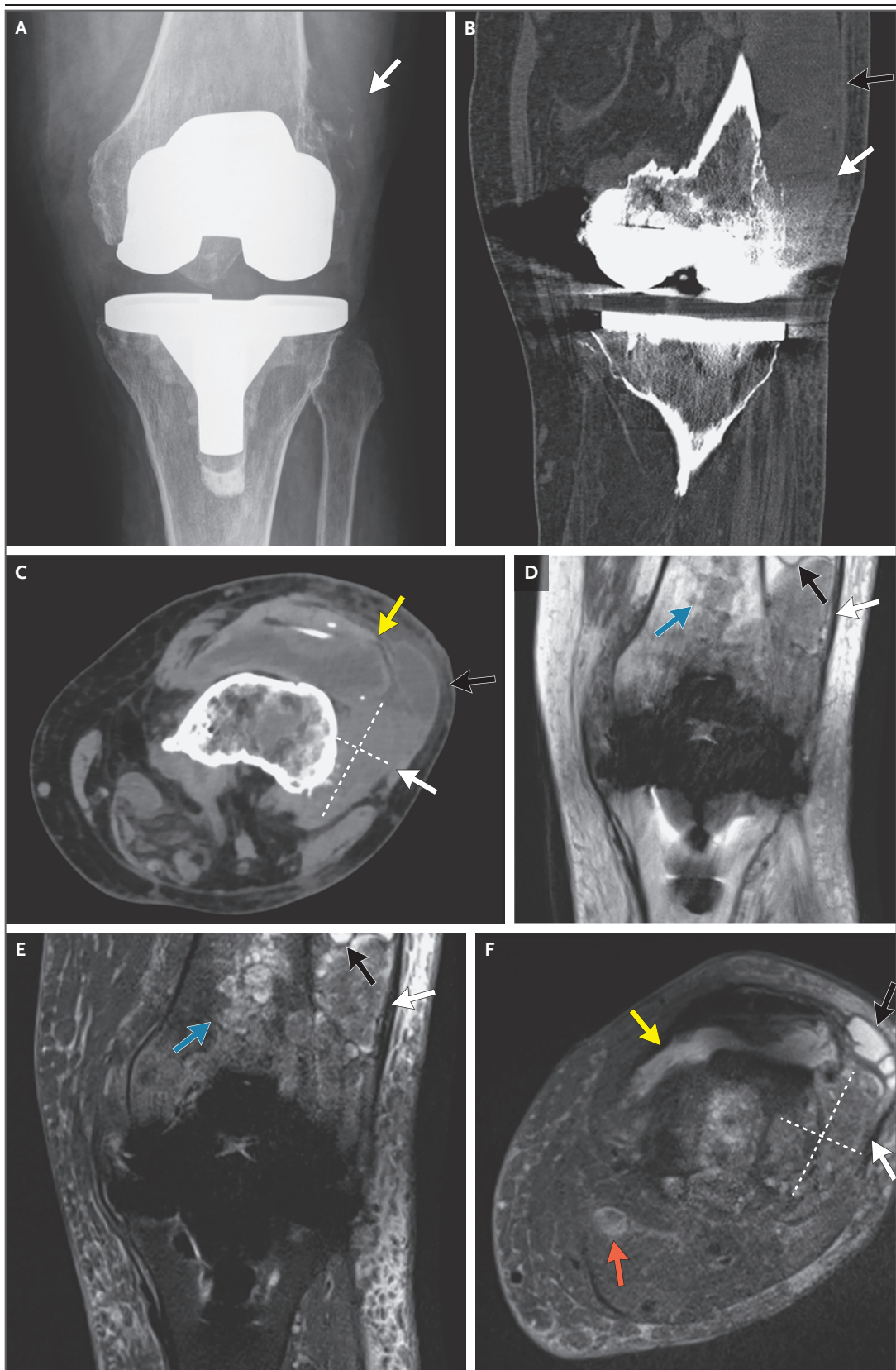


Figure 1 (facing page). Imaging Studies.

A radiograph of the left knee (Panel A) shows total knee arthroplasty with no periprosthetic fracture or lucency. Cortical erosions, osteolysis, and adjacent osseous fragmentation are present at the lateral aspect of the distal femoral metadiaphysis (Panel A, arrow). CT of the left knee was performed without the administration of intravenous contrast material. Coronal and axial images (Panels B and C, respectively) show a solid mass at the lateral aspect of the left knee, with underlying cortical remodeling, erosion, and osteolysis (Panels B and C, white arrows). The mass measures 4.9 cm by 3.0 cm (Panel C, dashed lines). An adjacent lobulated fluid collection is present (Panels B and C, black arrows), along with a joint effusion and synovitis (Panel C, yellow arrow). MRI of the left knee was also performed without the administration of intravenous contrast material. A coronal proton-density-weighted image (Panel D), a coronal short-tau inversion recovery (STIR) image (Panel E), and an axial STIR image (Panel F) show a mass that is adjacent to the lateral femoral condyle, with involvement of the lateral joint space, and has intermediate-to-low signal intensity (Panels D, E, and F, white arrows). The mass measures 6.0 cm in the greatest dimension (Panel F, dashed lines). An adjacent superficial, lobulated fluid collection is present (Panels D, E, and F, black arrows), extending into the soft tissue of the superolateral knee. A large joint effusion with synovitis (Panel F, yellow arrow) and a Baker's cyst (Panel F, red arrow) are present, along with heterogeneous red marrow in the distal femur (Panels D and E, blue arrows).

Dr. Shiang: Two days after the evaluation, computed tomography (CT) of the left knee (Fig. 1B and 1C), performed without the administration of intravenous contrast material, showed normal alignment of the components of the total knee arthroplasty and no hardware complications. A solid, masslike soft-tissue lesion (measuring 4.9 cm by 3.0 cm) was present at the lateral aspect of the distal femoral metadiaphysis, with underlying cortical remodeling, erosion, and osteolysis. An adjacent superficial, lobulated fluid collection was noted. Additional imaging was recommended.

Five days before the current presentation, magnetic resonance imaging (MRI) of the left knee (Fig. 1D, 1E, and 1F), performed without the administration of intravenous contrast material, showed the postoperative knee with a mass (measuring 6.0 cm in the greatest dimension) that was adjacent to the lateral femoral condyle, with involvement of the lateral joint space, and had intermediate-to-low signal intensity on short-tau inversion recovery (STIR) sequences. An ad-

jacent superficial, lobulated fluid collection was present, extending into the soft tissue of the superolateral knee. An associated large joint effusion and extensive synovitis were noted.

Dr. Yang: Bacterial cultures of synovial fluid showed no growth after 14 days of incubation. The patient was referred to the orthopedic surgery clinic of this hospital for further evaluation.

In the clinic, the patient reported ongoing left knee pain. He had received no glucocorticoid injections in the knee; there was no history of joint infection. He had undergone regular dental cleanings, with no dental caries or decay. He had a history of venous stasis in the legs, which had been complicated by stasis dermatitis and small, shallow ulcers. One year before the current presentation, a superficial swab of one of the ulcers grew methicillin-susceptible *Staphylococcus aureus*, for which the patient was treated with a 14-day course of cephalexin. Most recently, an ulcer on the distal aspect of the left leg was treated with a 3-month course of Unna boot therapy, which had been completed 7 months before the current presentation.

Additional medical history included hypertension, dyslipidemia, coronary artery disease, heart failure with preserved ejection fraction, hypothyroidism, benign prostatic hyperplasia, and sleep apnea, which was treated with continuous positive airway pressure at night. In addition to total knee arthroplasty, the surgical history included placement of an inferior vena cava filter because of gastrointestinal bleeding during anticoagulant therapy for provoked venous thromboembolism. Five years before the current presentation, the patient had received intravesical bacille Calmette–Guérin (BCG) for the treatment of bladder carcinoma. His last hospital admission had occurred at that time; records were not available, but the patient reported that it had taken place shortly after the tenth cycle of intravesical BCG and had involved treatment for sepsis in the intensive care unit.

Current medications included amlodipine, carvedilol, empagliflozin, finasteride, levothyroxine, simvastatin, tamsulosin, torsemide, and valsartan. Angiotensin-converting-enzyme inhibitors had caused angioedema of the tongue and lips. The patient was retired from work in sales. He was widowed and lived alone, without any pets, in an independent senior living community in Massachusetts. He had moved to Massachusetts from

Florida 5 years earlier and had not traveled recently. He drank alcohol occasionally; he had stopped smoking cigarettes 43 years before the current presentation and did not use illicit drugs. His family history included diabetes mellitus in his mother and coronary artery disease in his brother.

On examination, the blood pressure was 148/78 mm Hg, the pulse 71 beats per minute, the respiratory rate 20 breaths per minute, and the oxygen saturation 90% while the patient was breathing ambient air. The body-mass index (the weight in kilograms divided by the square of the height in meters) was 36.4. Other than bruising on the medial aspect of the left knee, at the site of the previous synovial-fluid aspiration, the examination was unchanged from that performed 3 weeks earlier. Additional synovial fluid was aspirated from the left knee. The fluid was bloody, with a total nucleated-cell count of 5657 per microliter, of which 79% were neutrophils, 11% lymphocytes, 7% monocytes, 2% eosinophils, and 1% macrophages. Gram's staining of synovial fluid showed rare polymorphonuclear leukocytes and no organisms; bacterial and fungal cultures of synovial fluid showed no growth.

A diagnostic test was performed.

DIFFERENTIAL DIAGNOSIS

Dr. Sandra B. Nelson: I am aware of the final diagnosis in this case. This 91-year-old man with a history of left total knee arthroplasty presented with left knee pain. Possible causes of pain after knee arthroplasty are numerous (Table 1) and may be intrinsic or extrinsic to the joint.¹ I will focus my discussion on three core features of this patient's presentation. First, the pain had a gradual onset and slow progression, occurring many years after successful knee arthroplasty. Second, a masslike soft-tissue lesion and fluid collection contiguous with the joint were present. Third, extensive synovitis was noted on imaging, and neutrophil-predominant leukocytosis was identified on synovial-fluid analysis.

NEOPLASM

Although cancer is not a common cause of pain after arthroplasty, the presence of a mass prompts consideration of a soft-tissue tumor, which may

Table 1. Causes of Pain after Knee Arthroplasty.*

Structural
Implant malpositioning
Implant failure
Implant loosening
Implant instability
Arthrofibrosis
Heterotopic ossification
Extensor dysfunction
Periprosthetic fracture
Extrinsic
Spine disease
Hip disease
Vascular disease
Tendonitis
Pes anserine bursitis
Inflammatory
Hemarthrosis
Inflammatory arthritis
Crystalline arthropathy
Particle disease
Periprosthetic joint infection

* Causes of pain are not mutually exclusive (e.g., infection may be associated with heterotopic ossification and implant loosening).

be either primary or metastatic. This patient had a history of bladder carcinoma. Osteoarticular metastasis can occur with bladder carcinoma but would typically develop after local invasion, which was not reported.

Primary soft-tissue sarcoma occurs rarely in older persons. The imaging features in this patient — including the intermediate-to-low signal intensity on STIR MRI and the associated fluid collection — make this diagnosis unlikely. Furthermore, sarcoma is typically painless and not associated with synovitis.

Tenosynovial giant-cell tumor, a benign clonal neoplasm that is characterized by hypervascular synovial proliferation, can be painful. However, this disease occurs primarily in younger adults and is not usually associated with tumorlike masses.² Tenosynovial giant-cell tumor occurring after arthroplasty is exceptionally rare and, when seen, is typically a recurrence. Although

a neoplasm cannot be ruled out on the basis of the available information, it would not sufficiently explain all the features of this patient's presentation.

HEMARTHROSIS

Hemarthrosis, or bleeding into the joint space, can be a cause of inflammatory synovitis. However, when hemarthrosis occurs years after arthroplasty, it is usually associated with trauma, anticoagulant therapy, or bleeding diathesis, none of which were reported in this case.

INFLAMMATORY ARTHRITIS

Inflammatory arthritis (e.g., rheumatoid arthritis) occurring after arthroplasty is rare, and it is typically a recurrence that develops in a patient with a preexisting diagnosis of this condition. New onset of inflammatory arthritis after total knee arthroplasty has been reported,³ but this patient had no other joint involvement. Furthermore, this diagnosis would be unusual in an older man.

CRYSTALLINE ARTHROPATHY

Although crystalline arthropathy involving native joints is common, crystalline arthropathy occurring after total knee arthroplasty is rare. In fact, a systematic review identified only 42 cases.⁴ The results of a crystal analysis of the synovial fluid were not provided for this patient, but the synovial-fluid characteristics would be compatible with crystalline arthritis. However, the subacute onset and chronic nature of the patient's pain and the presence of a juxtaarticular mass argue against this diagnosis.

PARTICLE DISEASE

Particle disease occurs when shedding of microscopic arthroplasty biomaterials causes an inflammatory response. Particle erosion is caused by wear at the device interface due to abrasion or stress fatigue. The femoral component and tibial baseplate used in total knee arthroplasty are typically constructed with metal such as cobalt-chromium or titanium alloy, whereas the tibial insert and often the patellar implant (if included) are made of highly cross-linked polyethylene (Fig. 2). Particle disease occurring after total knee arthroplasty is most likely to result from

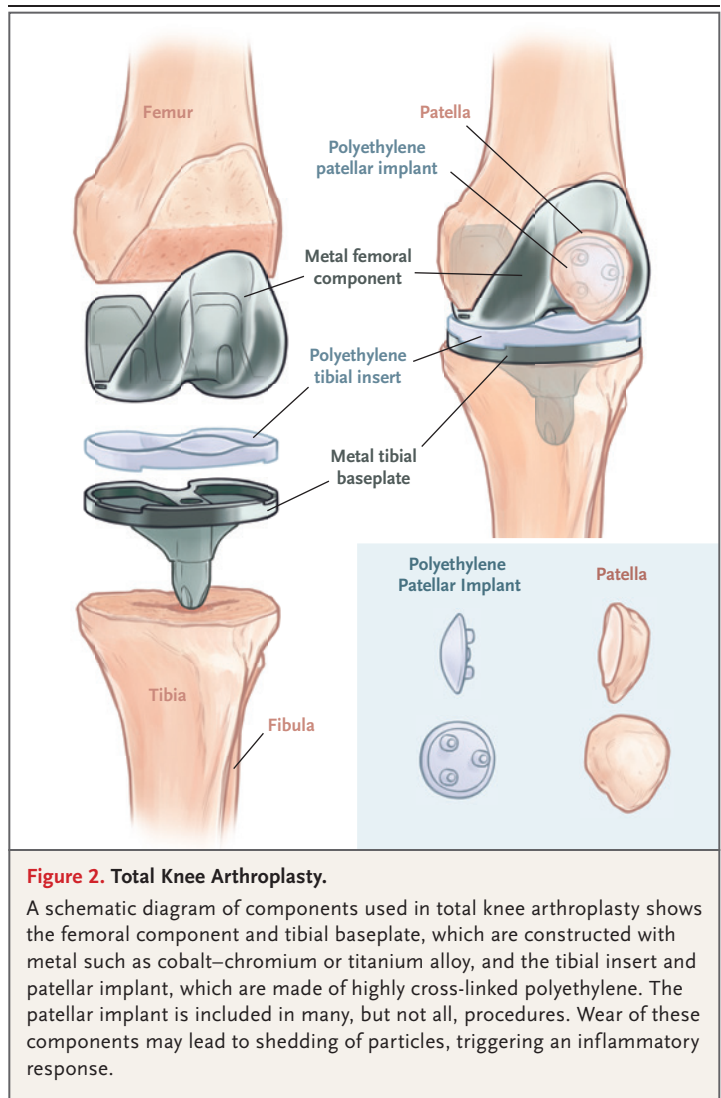


Figure 2. Total Knee Arthroplasty.

A schematic diagram of components used in total knee arthroplasty shows the femoral component and tibial baseplate, which are constructed with metal such as cobalt-chromium or titanium alloy, and the tibial insert and patellar implant, which are made of highly cross-linked polyethylene. The patellar implant is included in many, but not all, procedures. Wear of these components may lead to shedding of particles, triggering an inflammatory response.

polyethylene wear due to contact between the tibial insert and the opposing metal surface.

Shedding of polyethylene particles initiates an inflammatory response with activation of osteoclasts, which in turn leads to localized bone resorption (osteolysis) that can result in loosening of the implant.⁵ In the presence of polyethylene wear, the synovial fluid typically has monocyte predominance and is less inflammatory than that seen in this patient.⁶ Furthermore, a juxtaarticular mass is not typically seen with polyethylene wear. In contrast, the imaging findings and synovial-fluid analysis in this case could be consistent with an adverse local tissue reaction to metal;

however, this process would not occur after total knee arthroplasty in the absence of advanced polyethylene wear, given that the metal components otherwise do not directly articulate.

PERIPROSTHETIC JOINT INFECTION

Periprosthetic joint infection (PJI) complicates 2 to 3% of arthroplasty procedures. PJI may be acute or chronic and may occur at any time after arthroplasty.⁷ The diagnosis of chronic PJI can be especially challenging because the symptoms, examination findings, and imaging characteristics overlap with those seen with other, non-infectious diseases. PJI is characterized by the presence of microorganisms in periprosthetic tissues, and yet the sensitivity of synovial-fluid culture for the diagnosis is only 65 to 78%.⁸ Negative cultures can be explained by previous exposure to antibiotic therapy (although such exposure was not reported in this case); by the presence of fastidious organisms that do not grow readily in conventional culture media; by the residing of organisms in periprosthetic tissue or on the implant surface, where they may not shed into synovial fluid; or by the reduced replication of biofilm organisms involved in chronic infection, as compared with that of organisms involved in acute infection, which limits their detection in tests that rely on replication. It is important to note that this patient's negative synovial-fluid cultures cannot rule out PJI.

In the absence of a positive synovial-fluid culture, synovial-fluid analysis can be used to determine the likelihood of PJI. In a meta-analysis that was performed by a working group to establish an international consensus on the definition of PJI,⁹ the presence of either a leukocyte count in synovial fluid of 3000 per microliter or higher or a neutrophil percentage in leukocytes of 75% or more had at least 95% specificity for PJI. Results of testing for synovial-fluid biomarkers, including α -defensin, leukocyte esterase, C-reactive protein, and calprotectin, can also support a diagnosis of PJI. The biomarker α -defensin is an antimicrobial peptide produced by activated neutrophils and macrophages; a positive test in synovial fluid is highly specific for the diagnosis of PJI.¹⁰ Both α -defensin tests and cell counts in synovial fluid may have false positive results in the presence of an adverse local tissue reaction to

metal, crystalline arthropathy, or inflammatory arthritis. However, these diagnoses can be ruled out on clinical grounds, and therefore, the diagnosis of PJI can be confirmed in this patient.

For the orthopedist and the infectious diseases physician, confirmation of PJI is not enough. Management decisions hinge on identification of the infecting pathogen. Almost any microorganism can cause PJI.¹¹ Infection may result from direct inoculation (which usually occurs near the time of implantation), contiguous spread (e.g., spread from adjacent osteomyelitis), or hematogenous spread. In this patient, direct inoculation is unlikely because of the prolonged period since total knee arthroplasty; 20 years had elapsed since the procedure. Contiguous spread is a possible mechanism in this patient, given that he had had a venous ulcer in the leg with cellulitis. However, his ulcer was reported as being on the distal aspect of the leg, which makes contiguous spread unlikely. Hematogenous spread is caused by organisms that travel within the bloodstream and seed the periprosthetic joint, in a process similar to that seen with native joint septic arthritis. Hematogenous infection can occur at any time after arthroplasty, and although most cases are acute, some manifest indolently. Given the time since arthroplasty and the absence of known contiguous infection in this case, hematogenous inoculation is the most likely explanation for this patient's PJI.

Hematogenous infection implies antecedent bacteremia, which may be clinically silent. Bacteremia occurs when organisms breach the skin or a mucosal surface. Mucosal translocation may be caused by oropharyngeal injuries (e.g., dental procedures, infection, trauma, or gingivitis), loss of gut integrity (e.g., enterocolitis), or urinary tract infection. Organisms may also be introduced directly from the skin through vascular access devices or other breaks in skin integrity. Although clues to the identity of the pathogen are not always present, the patient history in this case reveals two possible causes of hematogenous infection.

First, this patient was reported to have had ulcer colonization with methicillin-susceptible *S. aureus* 1 year before the current presentation. *S. aureus* is the most common cause of hematogenous PJI.¹² Because of the intrinsic virulence of

S. aureus, hematogenous PJI resulting from *S. aureus* bacteremia typically has an acute and early onset; late onset of PJI after the development of *S. aureus* bacteremia is rare.^{13,14} This patient had chronic symptoms and did not have a systemic illness; these features make PJI with *S. aureus* unlikely. However, *S. aureus* infection should not be ruled out and cannot be discounted in this case.

Second, this patient had received intravesical BCG immunotherapy 5 years earlier. BCG was initially developed as a vaccine against *Mycobacterium tuberculosis* through attenuation of a closely related bovine strain, *M. bovis*. Its potency as an immunotherapeutic agent against cancer was recognized in the 1970s, and it is now standard care for select patients who have bladder cancer without muscle invasion. Administration of intravesical BCG may lead to active mycobacterial replication within host tissues,¹⁵ thereby inducing a local immune response against cancer.¹⁶ What constitutes infection after BCG administration is not clearly delineated, but it may occur in up to 4% of patients,¹⁷ manifesting as local genitourinary infection or disseminated disease. Rare cases of BCG sepsis have been reported.¹⁵

Similar to *M. tuberculosis*, BCG may establish latency. On reactivation, BCG infection typically has a subacute onset, which can include the development of inflammatory masses. Osteoarticular complications of BCG dissemination can include PJI, which typically occurs months to years after BCG administration.¹⁸⁻²¹ This patient was reported to have had sepsis after BCG administration. Although the cause of sepsis and the treatment approach are unknown, it is plausible that sepsis was due to BCG, which would suggest hematogenous seeding. Although PJI due to BCG is rare, this diagnosis matches all the core features in this case, including the time course and chronicity, the inflammatory synovitis, and the presence of a soft-tissue mass. This organism grows readily in mycobacterial culture, and in this case, the diagnosis of PJI due to *M. bovis* BCG could be confirmed with a mycobacterial culture of periprosthetic tissue, synovial fluid, or both.

DR. SANDRA B. NELSON'S DIAGNOSIS

Periprosthetic joint infection with *Mycobacterium bovis* bacille Calmette–Guérin.

SURGICAL MANAGEMENT

Dr. Erik T. Newman: The results of this patient's synovial-fluid analysis were consistent with PJI, and his symptoms and imaging findings were suggestive of implant loosening, which was presumably related to infection. Because the imaging findings were not suggestive of a neoplasm, I did not pursue needle biopsy preoperatively. I planned to perform intraoperative tissue sampling for frozen-section analysis, débridement and irrigation, and then a one-stage component revision.

I considered a two-stage approach, which would have involved a component explant and placement of a temporizing, joint-stabilizing spacer with antibiotic-impregnated cement, followed 2 to 3 months later by definitive reconstruction after completion of parenteral antimicrobial therapy. However, I decided against offering the two-stage approach. This older, frail patient with chronic foot drop would have struggled with the weight-bearing and mobility limitations imposed by spacer placement. In addition, soft-tissue healing in the presence of profound venous stasis would have proven increasingly challenging with repeat operations. Finally, the indolent nature of his presentation suggested that success was likely with a one-stage approach.

Under ultrasonographic guidance, the soft-tissue process at the lateral knee was sampled with a core needle. Frozen-section analysis showed chronic inflammation and necrotic debris, without evidence of cancer. Limited arthrotomy was performed. An area of white, fleshy soft tissue was seen arising from the lateral condyle of the femur. A generous tissue sample was taken, and frozen-section analysis showed histologic findings similar to those noted previously. Additional tissue was sent for microbiologic analysis. Empirical treatment with vancomycin and ceftriaxone was administered.

The knee was widely exposed. The inflammatory tissue at the lateral femur was extirpated, which resulted in a large bone defect. The femoral and tibial components were found to be visibly loose and readily removed; cement mantles were largely debonded, and residual cement was entirely removed. Aggressive synovectomy, débridement, and thorough irrigation were per-

formed. The limb was prepared again and draped, and a formal second surgical setup was used. The knee was reconstructed with long-stemmed, hinged implants and affixed with cement impregnated with vancomycin and gentamicin. The wound was closed over a drain, and the patient recovered uneventfully.

DIAGNOSTIC TESTING

Dr. Sarah M. Schrader: Tissue specimens were submitted for bacterial, fungal, and mycobacterial culture. Tissue was also frozen so that it could be saved for possible molecular testing. After 11 days of incubation, the cultures showed no growth, and the frozen specimen was sent for bacterial and mycobacterial DNA testing, performed with polymerase-chain-reaction amplification and sequencing. However, 3 days later, the mycobacterial culture showed growth in liquid medium. The microbiology laboratory performed Ziehl–Neelsen staining to confirm the presence of acid-fast bacilli, and the working diagnosis was updated to mycobacterial PJI.

Soon thereafter, the DNA test was found to be positive for sequences specific to the *M. tuberculosis* complex. In addition, the organism that had grown in the microbiology laboratory was identified as *M. tuberculosis* complex with the use of matrix-assisted laser desorption ionization–time-of-flight (MALDI-TOF) mass spectrometry, which identifies microorganisms on the basis of their unique protein signatures. The working diagnosis was updated to PJI due to *M. tuberculosis* complex. The *M. tuberculosis* complex comprises several species, including the attenuated *M. bovis* strain BCG. Neither the DNA test nor MALDI-TOF mass spectrometry allowed for discrimination of species within the complex.

Next, a nucleic acid test designed for species discrimination was performed. The test amplifies four genomic regions, the presence of which varies among species within the *M. tuberculosis* complex.²² A pan-*M. tuberculosis* complex sequence was present, and all four genomic regions targeted by the test were absent. This result confirmed the diagnosis of PJI due to *M. bovis* BCG.

The final diagnostic method was antimicrobial susceptibility testing. The isolate was first

tested with a molecular approach in which genes involved in resistance to antimycobacterial drugs were amplified and sequenced.²³ The isolate was predicted to be susceptible to isoniazid, rifampicin, and ethambutol. An H57D mutation in the gene *pncA* was detected, a finding consistent with resistance to pyrazinamide. This mutation is characteristic of *M. bovis*, which has intrinsic pyrazinamide resistance. Susceptibility to isoniazid, rifampicin, and ethambutol was later confirmed with a phenotypic assay. The isolate was unable to grow in liquid medium that contained these antibiotics, a finding indicating the absence of acquired drug resistance.

MICROBIOLOGIC DIAGNOSIS

Periprosthetic joint infection with drug-susceptible *Mycobacterium bovis* bacille Calmette–Guérin.

INFECTIOUS DISEASES MANAGEMENT AND FOLLOW-UP

Dr. Barbra M. Blair: A broad differential diagnosis for PJI was considered in this patient's case, including *M. bovis*. However, the most likely cause was initially thought to be a more common bacterial pathogen. Thus, while cultures of intraoperative samples were incubating, the patient received empirical treatment with daptomycin and ceftriaxone, which have activity against the most likely bacterial pathogens: staphylococcus, streptococcus, enterococcus, cutibacterium, and aerobic enterobacterales species.

When the intraoperative samples showed growth of acid-fast bacilli in liquid broth, *M. tuberculosis* complex was suspected. The diagnosis was confirmed within 48 hours by means of separate molecular testing performed on a tissue specimen. To evaluate for disseminated disease and additional sites of infection (e.g., the bladder or vasculature), imaging of the torso was performed, and blood and urine samples were obtained for mycobacterial culture. CT of the chest showed no abnormalities, and sputum was therefore not obtained for culture.

Treatment with daptomycin and ceftriaxone was discontinued. Although *M. bovis* BCG was the most suspected pathogen at this point, other *M. tuberculosis* complex species could not be ruled

out. Because the patient had no risk factors for the acquisition of multidrug-resistant *M. tuberculosis*, a typical regimen for empirical treatment — rifampin, isoniazid, ethambutol, and pyrazinamide — was considered. However, pyrazinamide can cause gout flares, and the patient was known to have an elevated uric acid level at baseline. Treatment with rifampin, isoniazid, ethambutol, and pyridoxine was started. Several of the patient's long-term medications were adjusted, given anticipated drug interactions with rifampin.

Three weeks after treatment was started, a generalized rash developed. The rash immediately resolved with discontinuation of antimycobacterial therapy. Subsequently, treatment was reintroduced, with one antimicrobial agent added per week. Treatment with isoniazid was started first, and no reaction was noted. One week after the addition of ethambutol, a pruritic urticarial eruption developed on the trunk. Treatment with isoniazid and ethambutol was stopped, and the rash resolved.

The patient expressed substantial distress about the antimycobacterial therapy and required laboratory monitoring, which he described as causing more toxic effects and disruptions than the infection itself. Extensive risk–benefit coun-

seling was provided, and the patient requested time to discuss the options with his children. Once the isolate was identified as *M. bovis* BCG, the patient was offered treatment with isoniazid and rifampin alone, which he declined.

Six months after surgery, the patient has healed well, without evidence of infection relapse or pain. He continues to live independently.

FINAL DIAGNOSIS

Periprosthetic joint infection with *Mycobacterium bovis* bacille Calmette–Guérin.

This case was presented at the Medicine Case Conference.

Disclosure forms provided by the authors are available with the full text of this article at NEJM.org.

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